

# CIS 371 Web Application Programming

## Pinia II

### App State Management



Lecturer: **Dr. Haoyu Li**



# Last time

What is state management?

What is Pinia?

- State,



# Practice: Pinia(States)

```
import { defineStore } from "pinia";
import products from "./products.json";
import { Product } from "./types";

export const useItemStore = defineStore("ItemStore", {
  state: () => {
    // Define and return the state object here.
    // It should include a list of items loaded from an external data file 'products.json'.
    return {
      items: products as Product[]
    };
  },
});
```

## Gamer's Delight Laptop

A powerful gaming laptop with top-tier graphics and performance for immersive gaming experiences.

**Price: \$1200.00**

★★★★☆ (4.0)

5 in stock

## SmartTech Pro Phone

Experience cutting-edge technology with our latest SmartTech Pro, featuring an ultra-responsive display and superior camera quality.

**Price: \$800.00**

★★★★★ (5.0)

10 in stock

## Rapid Dual USB-C Charger

Efficiently power up your devices with our compact, high-speed USB-C charging brick designed for the modern tech-savvy user.

**Price: \$12.00**

★★★★☆ (4.0)

20 in stock

Answer



# Pinia Actions

# Pinia Actions

```
import { defineStore } from "pinia";
import products from "../data/products.json";

export const useItemStore = defineStore("ItemStore", {
  state: () => {
    return { products };
  },

  actions: {
    fill() {},
  },
});
```

*ItemStore.ts*

Define actions as methods on the actions option

# Pinia Actions

```
import { defineStore } from "pinia";

export const useUserStore = defineStore("UserStore", {
  state: () => {
    return { users: [] };
  },

  actions: {
    async fill() {
      const res = await fetch(
        "https://randomuser.me/api?results=5&nat=gb,fr"
      );
      this.users = await res.json();
    },
  },
});
```

*UserStore.ts*

```
actions: {
  async fill() {
    const res = await fetch(
      "https://randomuser.me/api?results=5&nat=gb,fr&inc=name,email,picture"
    );
    this.users = await res.json();
    this.someOtherAction();
  },
  async someOtherAction() {},
},
});
```

Access state with `this`

Access other actions with `this`

# Pinia Actions

user.ts

```
type User = {  
  name: {  
    first: string;  
    last: string;  
    title: string;  
  };  
  email: string;  
  picture: {  
    large: string;  
    medium: string;  
    thumbnail: string;  
  };  
};  
type RandomUser = {  
  results: Array<User>;  
};  
export type { User, RandomUser };
```

UserStore.ts

```
import { defineStore } from "pinia";  
import { RandomUser } from "../types/user";  
  
export const useUserStore = defineStore("UserStore", {  
  state: () => {  
    return { users: { results: [] } as RandomUser };  
  },  
  
  actions: {  
    async fill() {  
      const res = await fetch(  
        "https://randomuser.me/api?results=5&nat=gb,fr&inc=name,email,picture"  
      );  
      this.users = await res.json();  
      console.log(this.users);  
    },  
  },  
});
```

# Pinia Actions

AnyVueComponent.vue

```
<script setup lang="ts">
import { useUserStore } from "../stores/UserStore";
const userStore = useUserStore();
userStore.fill();
</script>

<template>
  <ul v-for="(u, idx) in userStore.users.results" :key="idx" :u="u">
    <li>{{ u.name.first }}. {{ u.name.last }}</li>
  </ul>
</template>
```





# Pinia Actions

```
<script setup lang="ts">
import { useUserStore } from "../stores/UserStore";
const { fill } = useUserStore();
fill();
</script>
```

*de-structure*

```
<script setup lang="ts">
import { useUserStore } from "../stores/UserStore";
import { storeToRefs } from "pinia";
const { fill } = storeToRefs(useUserStore());
fill();
</script>
```

*won't work*



# Pinia Actions: Example

```
import { defineStore } from "pinia";

export const useCartStore = defineStore("CartStore", {
  state: () => {
    return {
      items: [],
    };
  },
  actions: {
    addItem(itemId, count) {
      // set the count for the proper item in the state above
    },
  },
});
```

CartStore.ts

```
<button @click="addItem(product.id, $event)">Add Product to Cart</button>
```

Item.vue

*Useful for updating state based on user interaction*

# Pinia Getters

# What are Pinia Getters?

Equivalent of computed props on a component

Must explicitly  
type the return

```
UserStore.ts

import { defineStore } from "pinia";
import { RandomUser } from "../types/user";

export const useUserStore = defineStore("UserStore", {
  state: () => {
    return { users: {} as RandomUser };
  },
  getters: {
    count(): number {
      return this.users.results.length;
    },
  },
  actions: {
    async fill() {
      const res = await fetch(
        "https://randomuser.me/api?results=5&nat=gb,fr&inc=name,email,picture"
      );
      this.users = await res.json();
    },
  },
});
```

Access state with `this`

# Pinia Getters

Access state with `state`

```
getters: {  
  count(state): number {  
    return state.users.results.length;  
  },  
},
```

No need to explicitly type return

```
getters: {  
  count: (state) => state.users.results.length,  
},
```

*single line arrow functions*

Access other  
getters on `this`

```
getters: {  
  count: (state) => state.users.results.length,  
  doubleCount(): number {  
    return this.count * 2;  
  },  
  findUserByFirstName: (state) => (first: string) => {  
    return state.users.results.find(  
      (user: User) => user.name.first === first  
    );  
  },  
},
```

Accept arguments by  
returning a function

# Access Getters

*as a property*

```
<script setup lang="ts">
import { useUserStore } from "../stores/UserStore";
const userStore = useUserStore();
console.log(userStore.count);
</script>
```

*de-structure*

```
<script setup lang="ts">
import { useUserStore } from "../stores/UserStore";
import { storeToRefs } from "pinia";
const { count } = storeToRefs(useUserStore());
console.log(count.value);
</script>
```

Demo

*Can de-structure getters from store but must use `storeToRefs`*



# Using Stores in Other Stores

CartStore.ts

```
import { defineStore } from "pinia";
import { useItemStore } from "../ItemStore";

export const useCartStore = defineStore("CartStore", {
  state: () => {
    return {
      items: [],
    };
  },
  getters: {
    allProducts() {
      const itemStore = useItemStore();
      return itemStore.products;
    },
  },
  actions: {
    addItem(itemId, count) {
      // set the count for the proper item in the state above
    },
  },
});
```

ItemStore.ts

```
import { defineStore } from "pinia";
import products from "../data/products.json";
export const useItemStore = defineStore("ItemStore", {
  state: () => {
    return { products };
  },
});
```

# Use actions from another store in an action

```
import { defineStore } from "pinia";
import { useItemStore } from "../ItemStore";

export const useCartStore = defineStore("CartStore", {
  state: () => {
    return {
      items: [],
    };
  },
  actions: {
    reloadProducts() {
      const itemStore = useItemStore();
      return itemStore.fill();
    },
  },
});
```

CartStore.ts

```
import { defineStore } from "pinia";
import products from "../data/products.json";
export const useItemStore = defineStore("ItemStore", {
  state: () => {
    return { products };
  },
  actions: {
    async fill() {
      //...
    },
  },
});
```

ItemStore.ts



# Use actions from another store in an action

CartStore.ts

```
import { defineStore } from "pinia";
import { useItemStore } from "../ItemStore";

const itemStore = useItemStore(); ❌
export const useCartStore = defineStore("CartStore", {
  state: () => {
    return {
      items: [],
    };
  },
  actions: {
    reloadProducts() {
      return itemStore.fill();
    },
  },
});
```



```
✖ >Uncaught Error: [🐞]: getActivePinia was called with no active Pinia. Did you
forget to install pinia?
    const pinia = createPinia()
    app.use(pinia)
    This will fail in production.
    at useStore (pinia.esm-browser.js:1638:19)
```

# Subscribe to Pinia Stores

Notify me when something changes in this store, so I can update the UI or perform other actions in response.

- Watch state for changes.
- Monitor actions for calls.
- Perform side effects.
- Measure how long your actions take to run.
- Trigger user notifications.
- Log errors to third-party services.

# Subscribe to actions

```
unsubscribe = userStore.$onAction(  
  ({  
    name, // name of the action  
    store, // store instance, same as `someStore`  
    args, // array of parameters passed to the action  
    after, // hook after the action returns or resolves  
    onError, // hook if the action throws or rejects  
  }) => {  
    // Action-related logic code here...  
  }  
);
```

*AnyVueComponent.vue*

# Use conditional to run on select actions

```
unsubscribe = userStore.$onAction(  
  ({  
    name, // name of the action  
    store, // store instance, same as `someStore`  
    args, // array of parameters passed to the action  
    after, // hook after the action returns or resolves  
    onError, // hook if the action throws or rejects  
  }) => {  
    if (name === "fill") {  
      // more code  
    }  
  }  
);
```

*AnyVueComponent.vue*



# After and onError

AnyVueComponent.vue

UserStore.ts

```
import { defineStore } from "pinia";
import { RandomUser } from "../types/user";

export const useUserStore = defineStore("UserStore", {
  state: () => {
    return { users: {} as RandomUser };
  },

  getters: {
    count: (state) => state.users.results.length,
  },
  actions: {
    async fill() {
      const res = await fetch(
        "https://randomuser.me/api?results=5&nat=gb,fr&inc=name,email,picture"
      );
      this.users = await res.json();
      return this.count;
    },
  },
});
```

```
unsubscribe = userStore.$onAction(
  ({
    name, // name of the action
    store, // store instance, same as `someStore`
    args, // array of parameters passed to the action
    after, // hook after the action returns or resolves
    onError, // hook if the action throws or rejects
  }) => {
    const startTime = Date.now();
    // this will trigger before an action on `store` is executed
    console.log(`Start "${name}" with params [${args.join(", ")}].`);
    if (name === "fill") {
      // after() will trigger if the action succeeds and after it has fully run.
      after((result) => {
        console.log(
          `Finished "${name}" after ${
            Date.now() - startTime
          }ms.\nResult: ${result}.`
        );
      });
    }
    // onError() will trigger if the action throws or returns a promise that rejects
    onError((error) => {
      console.warn(
        `Failed "${name}" after ${
          Date.now() - startTime
        }ms.\nError: ${error}.`
      );
    });
  });
}
```

Demo

# Subscribe to the state

*AnyVueComponent.vue*

```
onMounted(() => {  
  userStore.$subscribe((mutation, state) => {  
    // 'direct' | 'patch object' | 'patch function'  
    mutation.type;  
  
    // same as userStore.$id  
    mutation.storeId;  
  
    // only available with mutation.type === 'patch object'  
    mutation.payload; // patch object passed to $patch()  
  });  
});
```

[Demo](#)

[Online Doc](#)

# The 'Patch Object' in Pinia

The `$patch()` method is a convenient way to apply multiple state changes at once. It simplifies state management by allowing batch updates.

- When to use: Handling complex state structures or when multiple state properties need to be updated simultaneously.
- Value Setting:
  - For each key in the patch object, the value is the new value that you want to set for that corresponding state property.
  - The `$patch()` method ensures that these updates are reactive and efficiently managed by Vue's reactivity system.

# The 'Patch Object' in Pinia

```
import { defineStore } from "pinia";

export const useUserStore = defineStore("userStore", {
  state: () => ({
    name: "",
    age: 0,
    email: "",
  }),
  // actions, getters, etc.
});
```

UserStore.ts

[Demo](#)

[Online Doc](#)

```
<script setup lang="ts">
import { onMounted } from "vue";
import { useUserStore } from "../stores/UserStore";
const userStore = useUserStore();

onMounted(() => {
  userStore.$subscribe((mutation, state) => {
    if (mutation.type === "patch object") {
      console.log("Patch object:", mutation.payload);
    }
  });
  userStore.$patch({
    name: "John Doe",
    email: "john@example.com",
  });
});
</script>

<template>
  <ul>
    <li>{{ userStore.name }}</li>
    <li>{{ userStore.age }}</li>
    <li>{{ userStore.email }}</li>
  </ul>
</template>
```

AnyVueComponent.vue



# Exercise

Complete the store-app implementation with the following functionalities:

- Implement a CartStore using Pinia to temporarily store the user's selected items for checkout.
- Utilize the \$patch method to add items into cart.
- Displays all items currently in the cart.

# Persist data?

## Refresh-proof your Pinia Stores